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COLLEGE OF COMPUTER SCIENCE
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University Clubs LAN

CIS326 – IT Infrastructure Management | Final Report
Section 06 | Group 1
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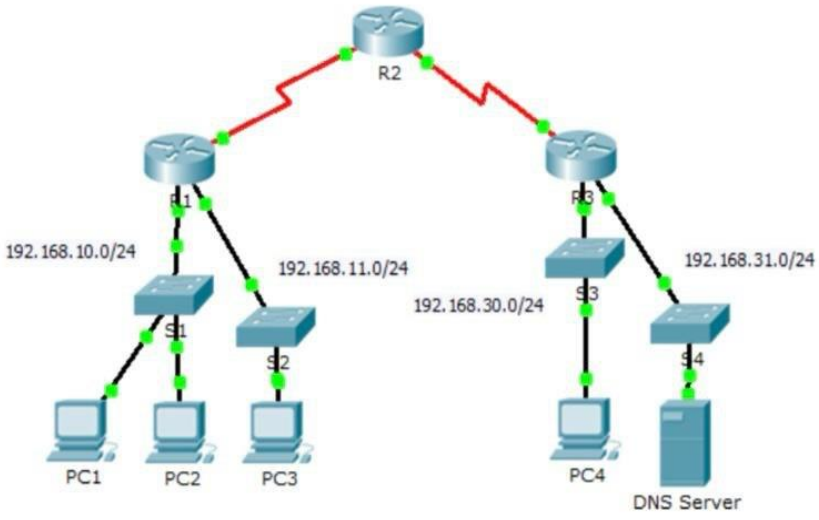
Introduction

In this report, we present the design and implementation of a structured network for the university student clubs using Cisco Packet Tracer. The project addresses communication and management inefficiencies between three primary entities: the Main Club Department, the Sports Club, and the Cultural Club. The inefficiencies were previously causing event organization delays and poor coordination. The suggested network entails wired and wireless connection via VLAN setup and centralized WLC, for enhanced segmentation, mobility, and control. Utilizing template 9 for its readability and , we developed a professional and functional topology that supports communication, simplifies administration, and supports the digitalization of student services.

Topology



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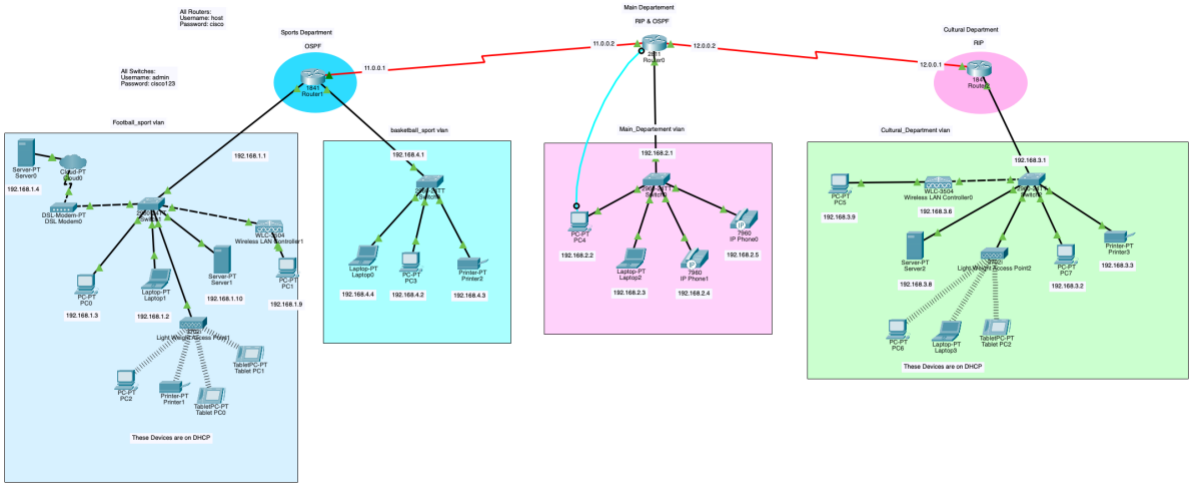
Requirements Of The Topology

Requirement	Quantity
Router	3
Switch	6
PCS-Laptops	9
Wireless LAN Controller	1
Servers	3
Printers	3
Access point	2
Cables	UTP, Serial, Console

Topology Set Up



Addressing Table

Device	Interface	IP address	Subnet Mask	Default Gateway	DNS Server	VLAN
Router 0	FastEthernet0/0	192.168.2.1	255.255.255.0	-	-	-
	Serial0/1/0	11.0.0.2	255.0.0.0	-	-	-
	Serial0/1/1	12.0.0.2	255.0.0.0	-	-	-
Router 1	FastEthernet0/0	192.168.1.1	255.255.255.0	-	-	-
	FastEthernet0/1	192.168.4.1	255.255.255.0	-	-	-
	Serial0/1/0	11.0.0.1	255.0.0.0	-	-	-
Router 2	FastEthernet0/0	192.168.3.1	255.255.255.0	-	-	-

	Serial0/1/1	12.0.0.1	255.0.0.0	-	-	-
PC0	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1	192.168.1.8	VLAN 201
PC1	FastEthernet0	192.168.1.9	255.255.255.0	192.168.1.1	-	VLAN 201
PC2	Wireless	192.168.1.13	255.255.255.0	192.168.1.1	192.168.1.8	VLAN 201
PC3	FastEthernet0	192.168.4.2	255.255.255.0	192.168.4.1	-	VLAN 202
PC4	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1	-	VLAN 100
PC5	FastEthernet0	192.168.3.9	255.255.255.0	192.168.3.1	-	VLAN 300
PC6	Wireless	192.168.3.17	255.255.255.0	192.168.3.1	192.168.3.7	VLAN 300
PC7	FastEthernet0	192.168.3.2	255.255.255.0	192.168.3.1	-	VLAN 300
Laptop 0	FastEthernet0	192.168.4.4	255.255.255.0	192.168.4.1	-	VLAN 202
Laptop1	FastEthernet0	192.168.1.5	255.255.255.0	192.168.1.1	192.168.1.8	VLAN 201
Laptop2	FastEthernet0	192.168.2.3	255.255.255.0	192.168.2.1	-	VLAN 100
Laptop3	Wireless	192.168.3.13	255.255.255.0	192.168.3.1	192.168.3.7	VLAN 300
Printer1	Wireless	192.168.1.6	255.255.255.0	192.168.1.1	192.168.1.8	VLAN 201
Printer2	FastEthernet0	192.168.4.3	255.255.255.0	192.168.4.1	-	VLAN 202
Printer3	FastEthernet0	192.168.3.3	255.255.255.0	192.168.3.1	-	VLAN 300
Tablet PC0	Wireless	192.168.1.8	255.255.255.0	192.168.1.1	192.168.1.8	VLAN 201
Tablet PC1	Wireless	169.254.226.107	255.255.0.0	-	-	VLAN 201
Tablet PC2	Wireless	192.168.3.16	255.255.255.0	192.168.3.1	192.168.3.7	VLAN 300
IP Phone 0	FastEthernet	192.168.2.2	-	192.168.2.1	-	VLAN 100
IP Phone 1	FastEthernet	192.168.2.5	-	192.168.2.1	-	VLAN 100

Server0	FastEthernet0	192.168.1.4	255.255.255.0	192.168.1.1	-	VLAN 201
Server1	FastEthernet0	192.168.1.10	255.255.255.0	192.168.1.1	-	VLAN 201
Server2	FastEthernet0	192.168.3.8	255.255.255.0	192.168.3.1	-	VLAN 300
WLC 0	Management	192.168.1.7	255.255.255.0	192.168.1.1	192.168.1.8	-
WLC 1	Management	192.168.3.6	255.255.255.0	192.168.3.1	192.168.3.7	-
LWAP 1	GigabitEthernet0	192.168.1.2	255.255.255.0	-	-	-
LWAP 2	GigabitEthernet0	192.168.3.10	255.255.255.0	-	-	-

Password Configuration

Commands used to apply a password to all routers:

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#username host password cisco
Router(config)#enable secret cisco
Router(config)#line console 0
Router(config-line)#password cisco
Router(config-line)#login
Router(config-line)#exit
Router(config)#line vty 0 4
Router(config-line)#password cisco
Router(config-line)#login
Router(config-line)#exit
Router(config)#
```

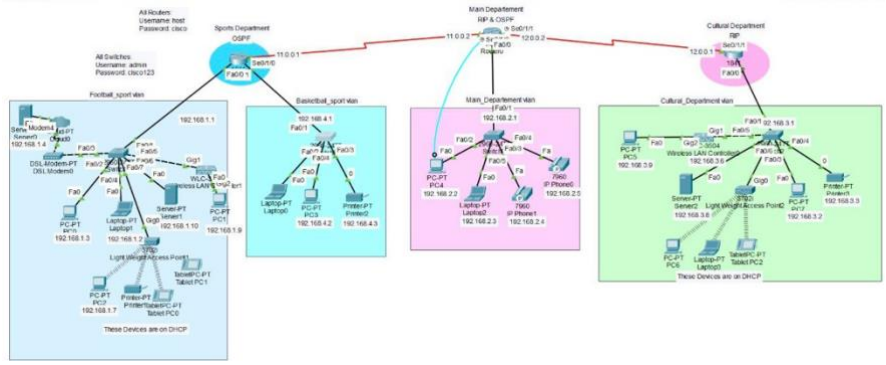
Commands used to apply a password to all switches:

```
SW0>enable
SW0#config t
Enter configuration commands, one per line.  End with CNTL/Z.
SW0(config)#username admin password cisco123
SW0(config)#line console 0
SW0(config-line)#password cisco123
SW0(config-line)#login
SW0(config-line)#exit
SW0(config)#line vty 0 4
SW0(config-line)#password cisco123
SW0(config-line)#login
SW0(config-line)#exit
SW0(config)#end
```

Password table:

Device	Password
Router 0	cisco
Router 1	cisco
Router 2	cisco
Switch 0	cisco123
Switch 1	cisco123
Switch 2	cisco123

VLAN Configuration



We have designed four separate VLANs in our network to organize and isolate traffic between departments:

- VLAN 201, named Football_sport, and VLAN 202, named Basketball_sport, are both connected to the Sports Department router.
- VLAN 100, named Main_Department, which is connected under the Main Department router.
- VLAN 300, named Cultural_Department, located under the Cultural Department router.

Create, name, and assign switch ports to the VLANs on the switches:

Switch 0:

```
SW0>enable
SW0#config t
Enter configuration commands, one per line.  End with CNTL/Z.
SW0(config)#vlan 100
SW0(config-vlan)#name Main_Department
SW0(config-vlan)#exit
SW0(config)#interface range fa0/2-5
SW0(config-if-range)#switchport mode access
SW0(config-if-range)#switchport access vlan 100
SW0(config-if-range)#exit
SW0(config)#exit
```

Switch 1:

```
SW1>enable
Password:
SW1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#vlan 201
SW1(config-vlan)#name Football_sport
SW1(config-vlan)#exit
SW1(config)#interface range fa0/2-7
SW1(config-if-range)#switchport mode access
SW1(config-if-range)#switchport access vlan 201
SW1(config-if-range)#exit
SW1(config)#exit
```

Switch 2:

```
SW2>enable
SW2#config t
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#vlan 300
SW2(config-vlan)#name Cultural_Department
SW2(config-vlan)#exit
SW2(config)#interface range fa0/2-6
SW2(config-if-range)#switchport mode access
SW2(config-if-range)#switchport access vlan 300
SW2(config-if-range)#exit
SW2(config)#exit
```

Switch 3:

```
SW3>enable
Password:
SW3#config t
Enter configuration commands, one per line.  End with CNTL/Z.
SW3(config)#vlan 202
SW3(config-vlan)#name Basketball_sport
SW3(config-vlan)#exit
SW3(config)#interface range fa0/2-4
SW3(config-if-range)#switchport mode access
SW3(config-if-range)#switchport access vlan 202
SW3(config-if-range)#exit
SW3(config)#exit
```

By using the “show vlan brief “ command, we can quickly see all the VLANs on the switch, their IDs, names, and which ports are part of each VLAN.

Switch 0:

```
SW0#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
100  Main_Departement                 active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Switch 1:

```
SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
201  Football_sport                   active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1#
```

Switch 2:

```
SW2#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
300  Cultural_Department              active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Switch 3:

```
SW3#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
202  Basketball_sport                 active    Fa0/2, Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

We apply Inter-VLAN routing on all the routers to allow communication between devices that are in different VLANs.

VLAN 100 Router 0 :

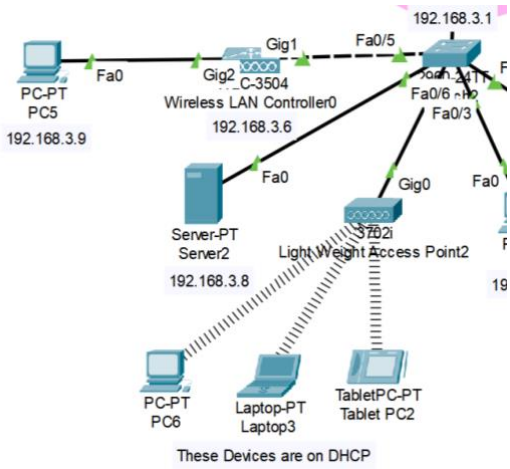
```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface f0/0.100
Router(config-subif)#encapsulation dot1Q 100
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#end
```

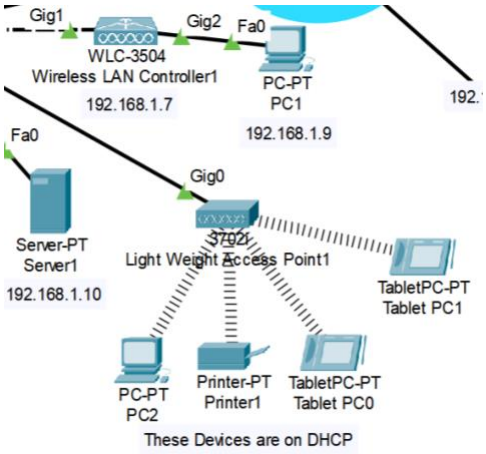
VLAN 201 & 202 Router 1:

```
Router>enable
Password:
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface f0/0.201
Router(config-subif)#encapsulation dot1Q 201
Router(config-subif)#ip address 192.168.1.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface f0/1.202
Router(config-subif)#encapsulation dot1Q 202
Router(config-subif)#ip address 192.168.4.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
```

VLAN 300 Router 2:

```
Router>enable
Password:
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface f0/0.300
Router(config-subif)#encapsulation dot1Q 300
Router(config-subif)#ip address 192.168.3.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
```

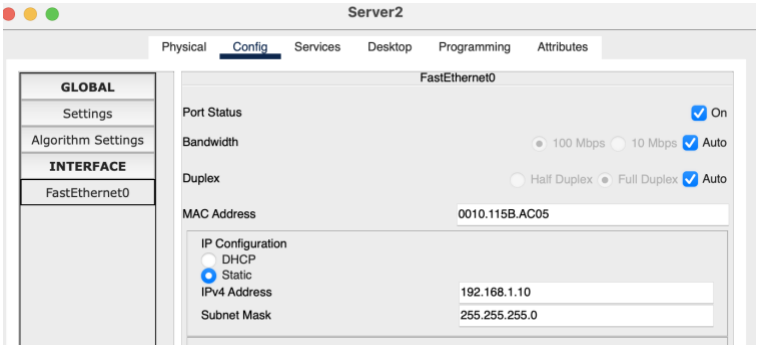




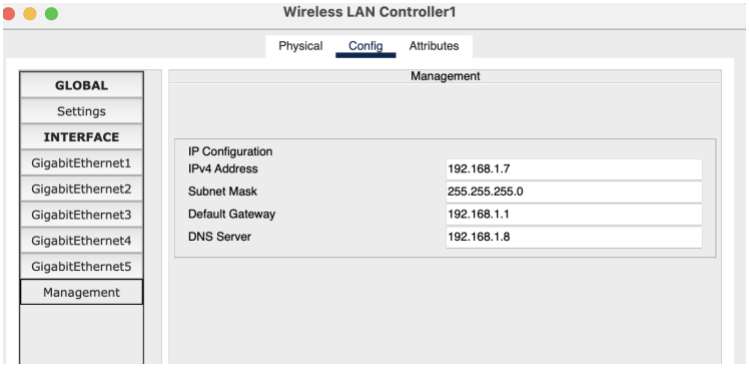
WLC And VOIP Configuration

1. Devices Configuration (Server, WLC, PC, Printer, Tablet, AP, VoIP, Modem and cloud):

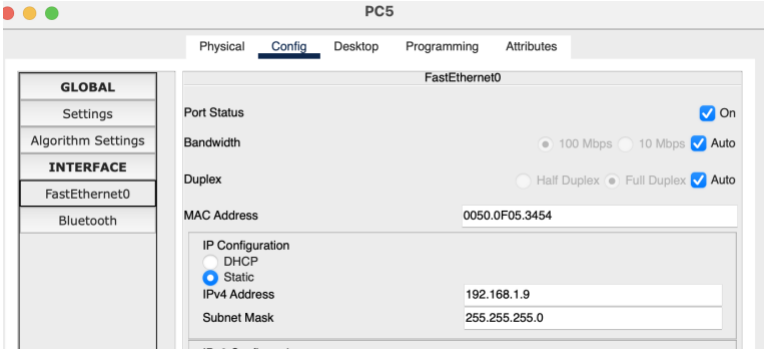
- Server:



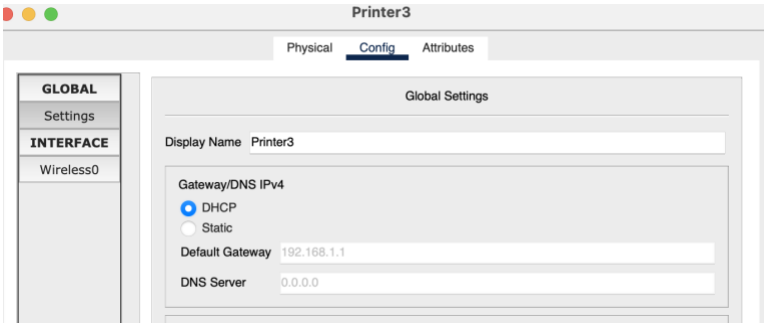
- WLC:



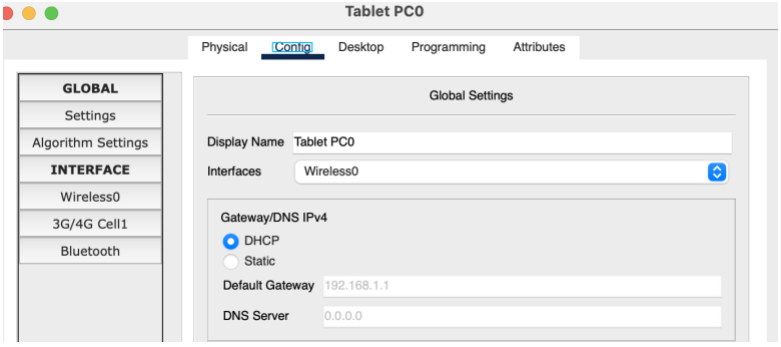
- PC:



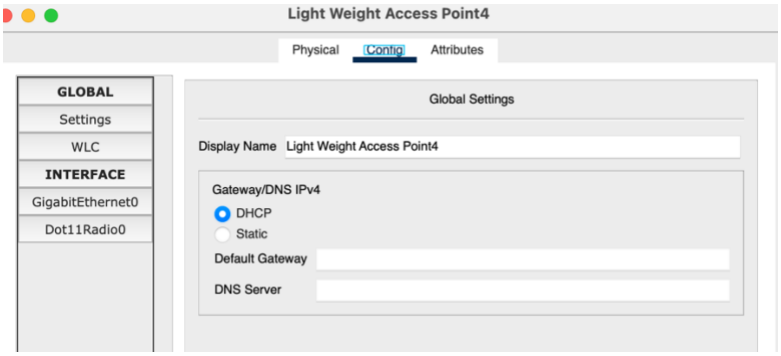
- Printer:



- Tablet:



- Access Point:



- VoIP Configuration:

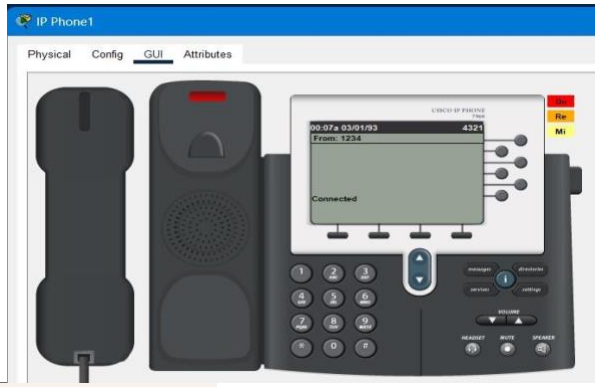
Configure interface on Router0:

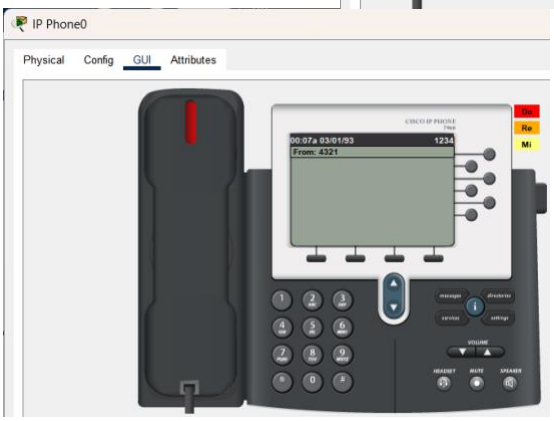
```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#exit
Router(config)#ip dhcp pool voip
Router(dhcp-config)#network 192.168.2.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.2.1
Router(dhcp-config)#option 150 ip 192.168.2.1
Router(dhcp-config)#exit
Router(config)#telephony-service
Router(config-telephony)#max-ephones 3
Router(config-telephony)#max-dn 3
Router(config-telephony)#ip source address 192.168.2.1 port 2000
                                         ^
% Invalid input detected at '^' marker.

Router(config-telephony)#ip source-address 192.168.2.1 port 2000
Router(config-telephony)#auto assign 4 to 6
Router(config-telephony)#auto assign 1 to 5
Router(config-telephony)#exit
Router(config)#telephony-service
Router(config-telephony)#ephone-dn 1
Router(config-ephone-dn)#%LINK-3-UPDOWN: Interface ephone_dsp DN 1.1, changed state to up

Router(config-ephone-dn)#number 1234
Router(config-ephone-dn)#ephone-dn 2
Router(config-ephone-dn)#%LINK-3-UPDOWN: Interface ephone_dsp DN 2.1, changed state to up

Router(config-ephone-dn)#4321
                         ^
% Invalid input detected at '^' marker.

Router(config-ephone-dn)#number 4321
Router(config-ephone-dn)#exit
Router(config)#
```

- configure on switch0:

```
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface range f0/1-15
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport voice vlan 1
Switch(config-if-range)#exit
Switch(config)#
```

- Check Connectivity:

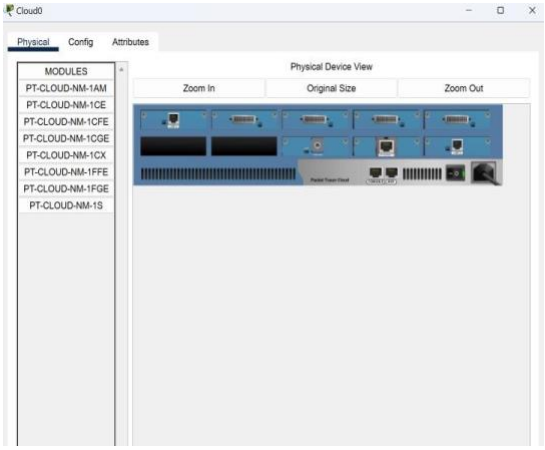


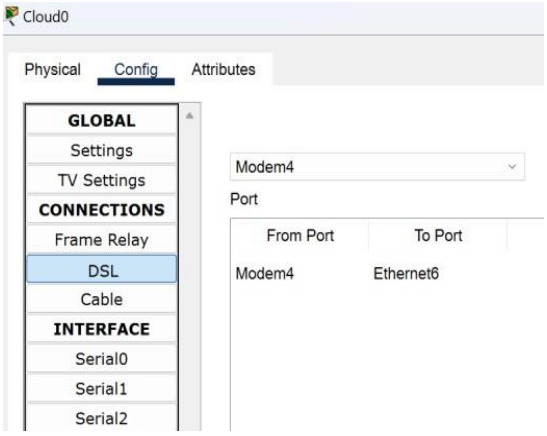


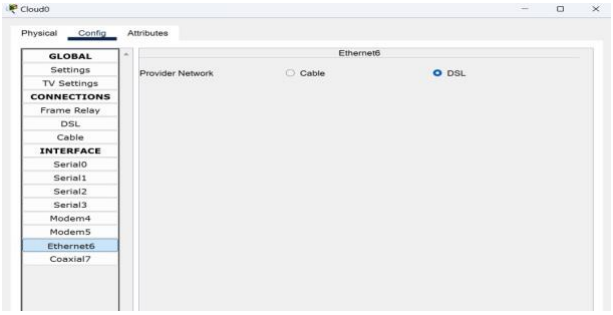
```
Device Name: IP Phone0
Device Model: 7960

Port    Link   IP Address         MAC Address
Vlan1   Up     192.168.2.4/24     000C.CF37.5B80
Switch  Up     <not set>          000D.BD65.E101
PC      Down   <not set>          000D.BD65.E102

Gateway: 192.168.2.1
Line Number: 1234

Physical Location: Intercity > Home City > Corporate Office > IP Phone
```

```
Device Name: IP Phone1
Device Model: 7960

Port    Link   IP Address         MAC Address
Vlan1   Up     192.168.2.5/24     00E0.B0A9.2D72
Switch  Up     <not set>          0001.644B.6501
PC      Down   <not set>          0001.644B.6502

Gateway: 192.168.2.1
Line Number: 4321

Physical Location: Intercity > Home City > Corporate Office > IP Phone1
```

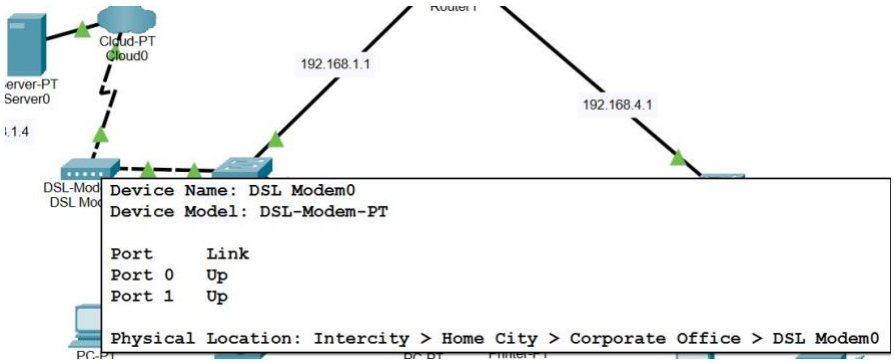
- Modem and Cloud configuration: connecting the cloud and the modem.





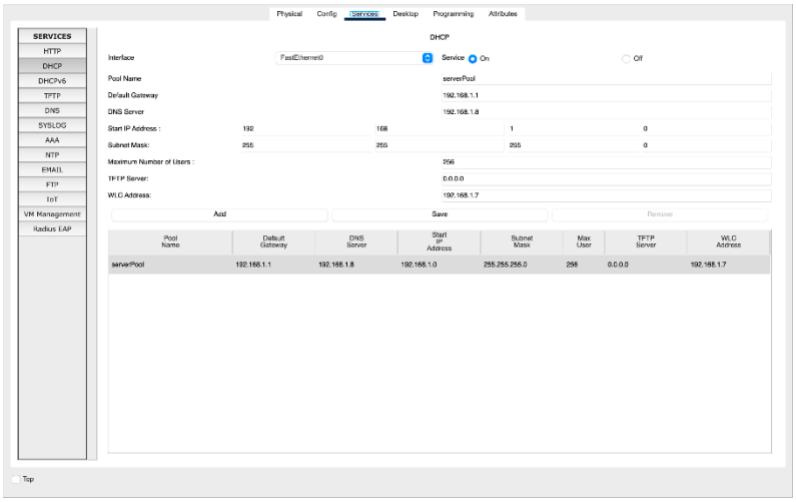


- check activation:



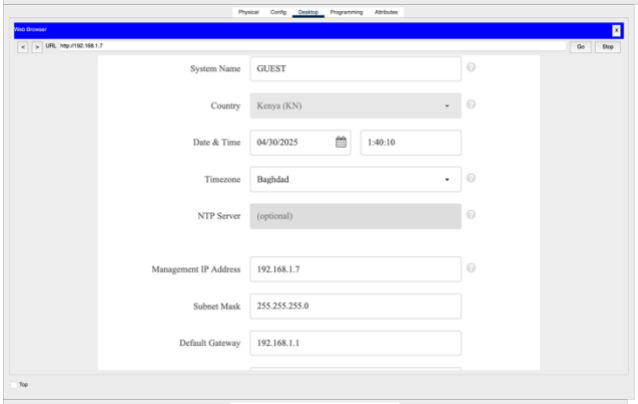
2. WLC Process:

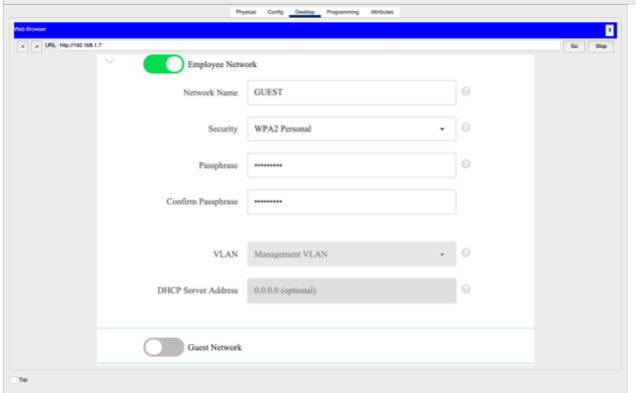
- Server:

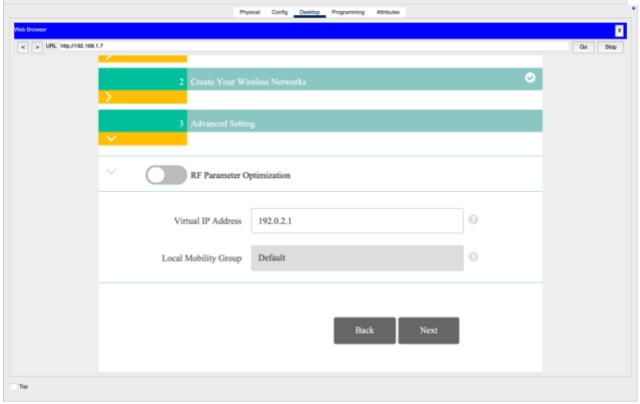


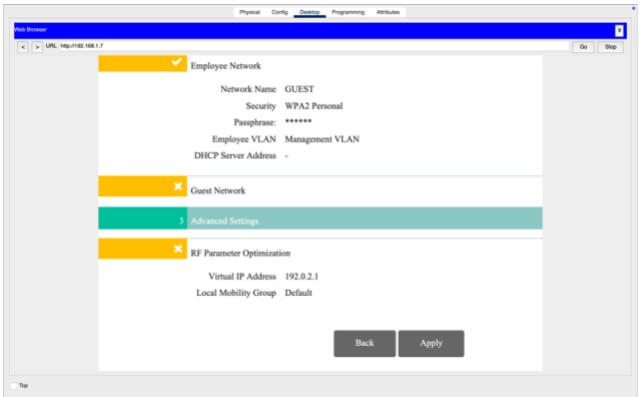
- PC:

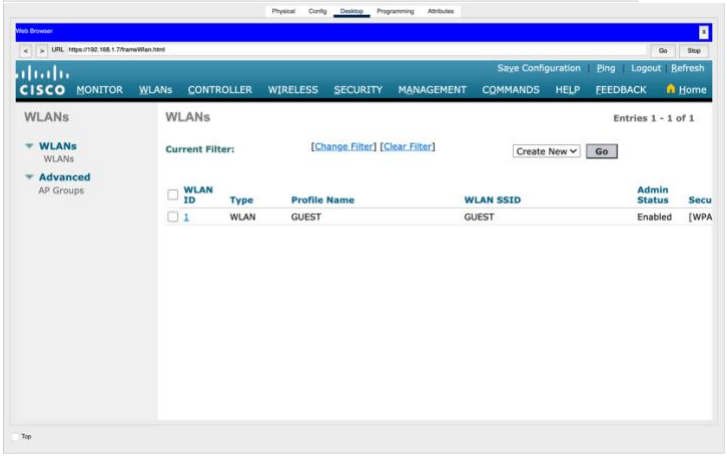
Cisco 3500 Series Wireless LAN Controller
Welcome! Please start by creating an admin account.
admin1
Start


System Name
GUEST
Country
Kenya (KN)
Date & Time
04/30/2025
1:40:10
Timezone
Baghdad
NTP Server
(optional)
Management IP Address
192.168.1.7
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1


Employee Network
Network Name
GUEST
Security
WPA2 Personal
Passphrase
Confirm Passphrase
VLAN
Management VLAN
DHCP Server Address
0.0.0.0 (optional)
Guest Network

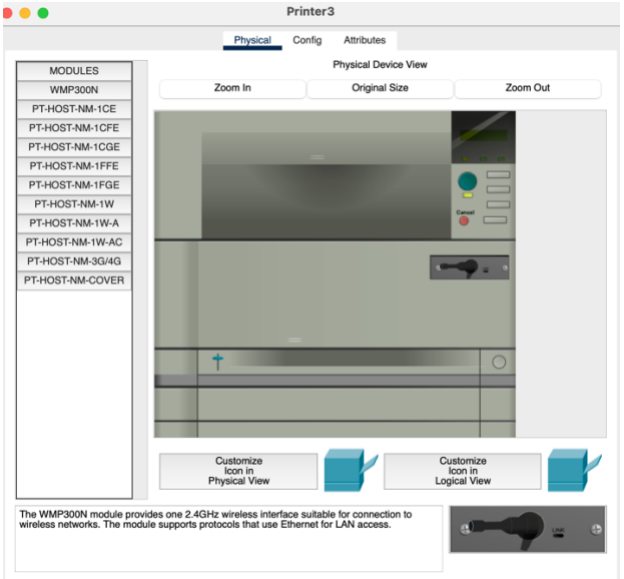

2 Create Your Wireless Networks
3 Advanced Setting
RF Parameter Optimization
Virtual IP Address
192.0.2.1
Local Mobility Group
Default
Back
Next

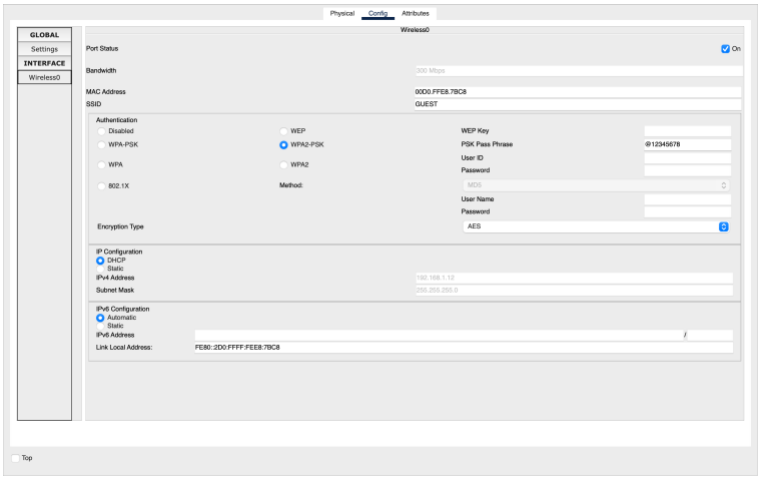




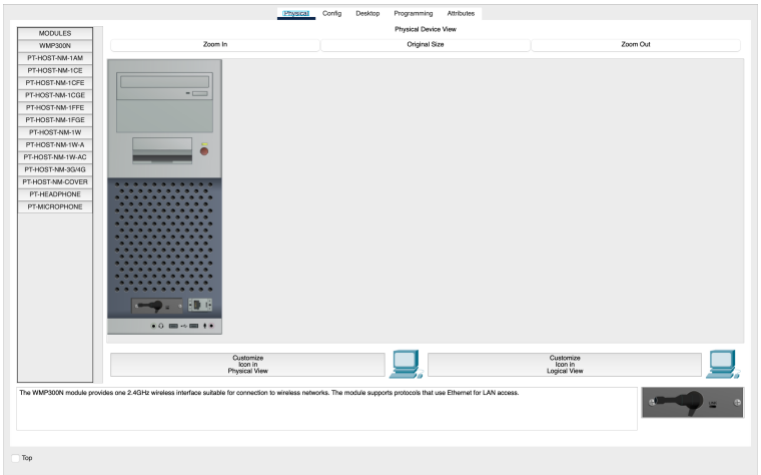
3. Connecting The Devices:

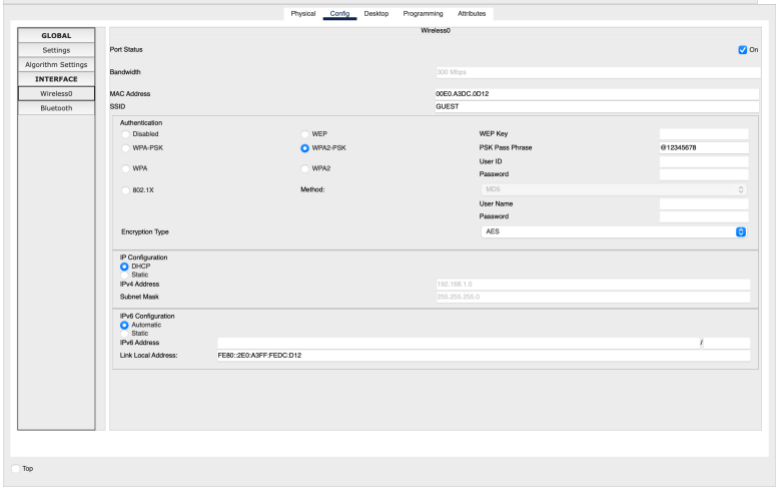
- Printer:


Printer3
Physical
Config
Attributes
Physical Device View
MODULES
WMP300N
PT-HOST-NM-1CE
PT-HOST-NM-1CFE
PT-HOST-NM-1CGE
PT-HOST-NM-1FFE
PT-HOST-NM-1FGE
PT-HOST-NM-1W
PT-HOST-NM-1W-A
PT-HOST-NM-1W-AC
PT-HOST-NM-3G/4G
PT-HOST-NM-COVER
Zoom In
Original Size
Zoom Out
Customize Icon in Physical View
Customize Icon in Logical View
The WMP300N module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module supports protocols that use Ethernet for LAN access.

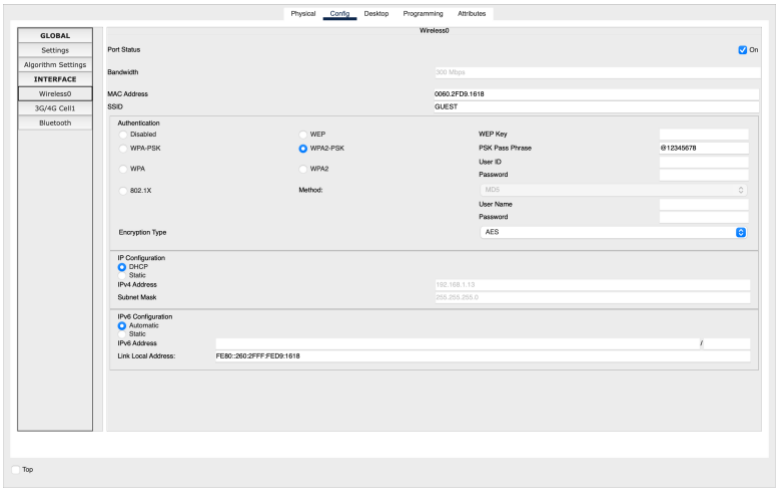

Physical
Config
Attributes
Wireless0
GLOBAL
Settings
INTERFACE
Wireless0
Port Status
On
Bandwidth
300 Mbps
MAC Address
00D0.FFE8.78C8
SSID
GUEST
Authentication
Disabled
WEP
WPA-PSK
WPA2-PSK
WPA
WPA2
802.1X
Method:
MD5
WEP Key
PSK Pass Phrase
@12345678
User ID
Password
User Name
Password
Encryption Type
AES
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.12
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::2D0:FFFF:FEE8:78C8
Top

- PC:


Physical
Config
Desktop
Programming
Attributes
Physical Device View
Zoom In
Original Size
Zoom Out
MODULES
WMP300N
PT-HOST-NM-1AM
PT-HOST-NM-1CE
PT-HOST-NM-1CFE
PT-HOST-NM-1CGE
PT-HOST-NM-1FFE
PT-HOST-NM-1FGE
PT-HOST-NM-1W
PT-HOST-NM-1W-A
PT-HOST-NM-1W-AC
PT-HOST-NM-3G/4G
PT-HOST-NM-COVER
PT-HEADPHONE
PT-MICROPHONE
Customize Icon in Physical View
Customize Icon in Logical View
The WMP300N module provides one 2.4GHz wireless interface suitable for connection to wireless networks. The module supports protocols that use Ethernet for LAN access.
Top


Physical
Config
Desktop
Programming
Attributes
Wireless0
GLOBAL
Settings
Algorithm Settings
INTERFACE
Wireless0
Bluetooth
Port Status
On
Bandwidth
300 Mbps
MAC Address
00E0.A3DC.0D12
SSID
GUEST
Authentication
Disabled
WEP
WPA-PSK
WPA2-PSK
WPA
WPA2
802.1X
Method:
MD5
WEP Key
PSK Pass Phrase
@12345678
User ID
Password
User Name
Password
Encryption Type
AES
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.5
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::2E0:A3FF:FEDC:D12
Top

- Tablet:


Physical
Config
Desktop
Programming
Attributes
Wireless0
GLOBAL
Settings
Algorithm Settings
INTERFACE
Wireless0
3G/4G Cell1
Bluetooth
Port Status
On
Bandwidth
300 Mbps
MAC Address
0060.2FD9.1618
SSID
GUEST
Authentication
Disabled
WEP
WPA-PSK
WPA2-PSK
WPA
WPA2
802.1X
Method:
MD5
WEP Key
PSK Pass Phrase
@12345678
User ID
Password
User Name
Password
Encryption Type
AES
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.13
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::260:2FFF:FED9:1618
Top

Configure Router interfaces

Router 0:

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface f0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#interface s0/1/0
Router(config-if)#ip address 11.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#interface s0/1/1
Router(config-if)#ip address 12.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 1:

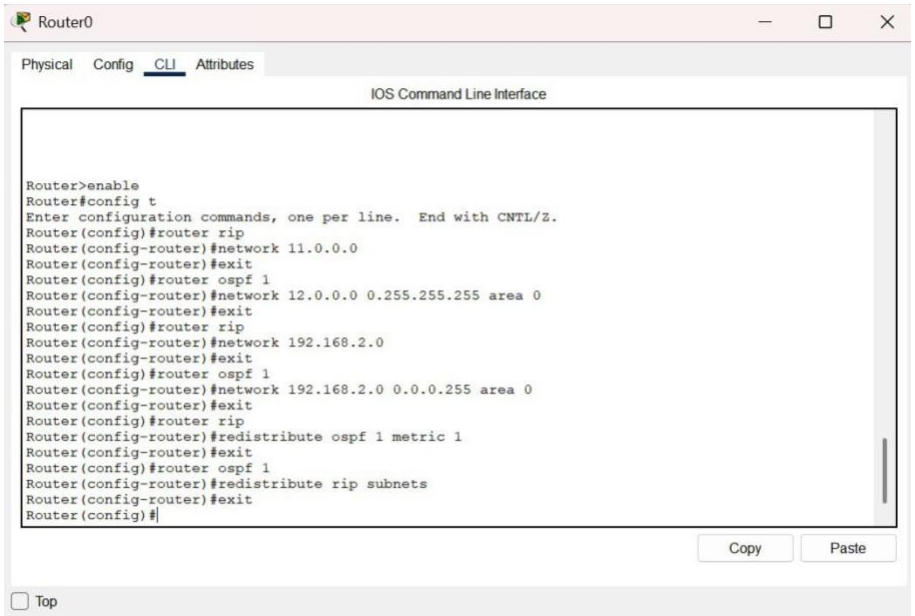
```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface f0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#interface f0/1
Router(config-if)#ip address 192.168.4.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#interface s0/1/0
Router(config-if)#ip address 11.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 2:

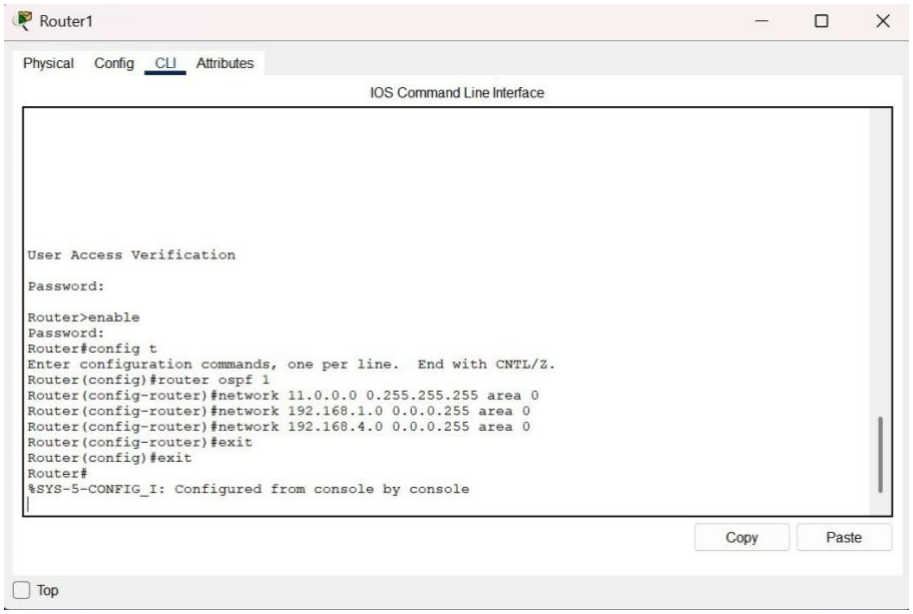
```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface f0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#interface s0/1/1
Router(config-if)#ip address 12.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Configure RIPV2 & OSPF Routing

Router 0 is used and configured to redistribute the routing to two routing protocols (RIP & OSPF) using these commands:



OSPF protocol is configured in Router 1 using these commands:



RIP protocol is configured in Router 2 using these commands:

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
User Access Verification

Password:

Router>enable
Password:
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 12.0.0.0
Router(config-router)#network 192.168.3.0
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

Top

Verifying Routing Protocols

The “show ip route” command is used to display the routing table on all routers:

Router 0:

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
User Access Verification

Password:

Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       11.0.0.0/8 is directly connected, Serial0/1/0
L       11.0.0.2/32 is directly connected, Serial0/1/0
     12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       12.0.0.0/8 is directly connected, Serial0/1/1
L       12.0.0.2/32 is directly connected, Serial0/1/1
S    192.168.1.0/24 [1/0] via 11.0.0.1
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, FastEthernet0/0
L       192.168.2.1/32 is directly connected, FastEthernet0/0
S    192.168.3.0/24 [1/0] via 12.0.0.1
S    192.168.4.0/24 [1/0] via 11.0.0.1

Router#
```

Router 1:

Router1

Physical Config CLI Attributes

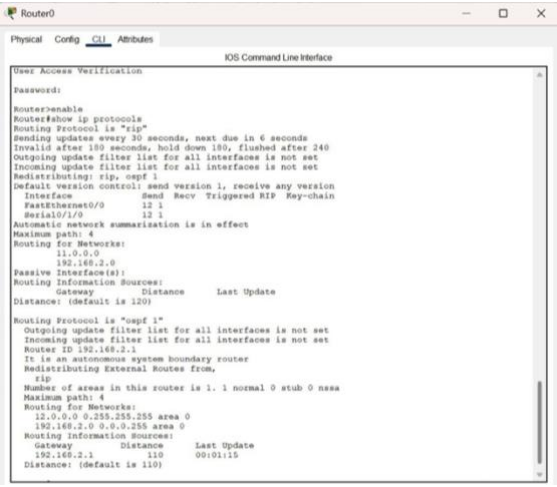
IOS Command Line Interface

```
User Access Verification

Password:

Router>enable
Password:
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    11.0.0.0/8 is directly connected, Serial0/1/0
S    12.0.0.0/8 [1/0] via 11.0.0.2
C    192.168.1.0/24 is directly connected, FastEthernet0/0
S    192.168.2.0/24 [1/0] via 11.0.0.2
S    192.168.3.0/24 [1/0] via 11.0.0.2
C    192.168.4.0/24 is directly connected, FastEthernet0/1

Router#
```

Router 2:

The “show is protocols” command is a quick way to verify vital OSPF & RIP information:

Router 0:



Router 1:

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
User Access Verification

Password:

Router>enable
Password:
Router#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.4.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    11.0.0.0 0.255.255.255 area 0
    192.168.1.0 0.0.0.255 area 0
    192.168.4.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    192.168.4.1          110      00:02:43
  Distance: (default is 110)

Router#
```

Router 2:

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Password:

Router>enable
Password:
Password:
Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 12 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 1, receive any version
  Interface             Send  Recv  Triggered RIP  Key-chain
  FastEthernet0/0       12 1
  Serial0/1/1           12 1
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
        12.0.0.0
        192.168.3.0
Passive Interface(s):
Routing Information Sources:
        Gateway         Distance      Last Update
Distance: (default is 120)
Router#
```

Testing pings

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC3	PC4	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC3	PC5	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC0	PC2	ICMP		0.000	N	2	(edit)	(delete)

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC4	PC6	ICMP		0.000	N	3	(edit)	(delete)
	Successful	PC2	Tablet PC0	ICMP		0.000	N	4	(edit)	(delete)
	Successful	PC2	PC3	ICMP		0.000	N	5	(edit)	(delete)

Case Study (Google Cloud)

1. Concept of Cloud Computing

Cloud computing is the delivery of computing resources such as servers, storage, networking, software, and analytics over the internet. Instead of relying on local servers or personal devices to run applications and store data, organizations can access these services from cloud providers on a pay-as-you-go basis. This model supports flexibility, scalability, and cost-efficiency.

Google Cloud Platform (GCP) is one of the world's top cloud service providers. It offers a vast array of tools and services to build, deploy, and manage applications in a highly secure and scalable environment. GCP runs on the same infrastructure that powers Google's own applications, such as Google Search, Gmail, and YouTube, ensuring users benefit from high performance, reliability, and global reach. (Google Cloud, 'What is Cloud Computing?', 2025) [1].

2. Cloud Computing Models and Usage Areas

Cloud services are offered under three main models:

- **Infrastructure as a Service (IaaS)**: IaaS offers virtual computing resources online. On GCP, this includes Compute Engine, which allows users to run virtual machines with full control over the OS and software—ideal for IT teams managing their own environments (Google Cloud, 'What is IaaS?', 2025) [4].

- **Platform as a Service (PaaS):** PaaS provides a ready-made platform for developers to build and deploy applications without handling infrastructure. GCP's App Engine manages scaling, load balancing, and monitoring, letting developers focus on coding (Google Cloud, 'Google Cloud Overview', 2025) [2].

- **Software as a Service (Saa**S): SaaS delivers complete software solutions via the internet. Google Workspace, including Gmail, Docs, and Drive, is a key example—requiring no setup or maintenance by users (Google Cloud, 'Google Cloud Overview', 2025) [2].

Usage Areas of Google Cloud:

Google Cloud supports a wide range of industries and technological needs:

- **Web and Application Hosting:**
  Businesses can use GCP to host websites, mobile apps, or backend services. Services like App Engine and Cloud Run offer serverless hosting, which automatically scales based on user demand and reduces infrastructure overhead (eG Innovations, 'Top Google Cloud Platform (GCP) Services Explained with Use Cases', 2025) [5].

- **Big Data Analytics and Business Intelligence**:
  With tools like BigQuery and Data Studio, organizations can analyze vast volumes of data in real time. These tools are used in sectors such as finance, retail, and healthcare to generate insights (eG Innovations, 'Top Google Cloud Platform (GCP) Services Explained with Use Cases', 2025) [5].

- **Machine Learning and Artificial Intelligence**:
  GCP provides powerful machine learning services that allow businesses to build custom models for tasks like image recognition, natural language processing, fraud detection, and predictive analytics (Google Cloud, 'Google Cloud Overview', 2025) [2].

- **Cloud Storage and Data Backup**:
  Google Cloud Storage provides secure, durable, and scalable storage solutions. It is commonly used for archiving, backup, disaster recovery, and serving static content (Google Cloud, 'Google Cloud Platform Services Summary', 2025) [7].

- **IoT (Internet of Things):**
  GCP offers services to connect, manage, and analyze data from millions of IoT devices. This is valuable for industries such as manufacturing, logistics, and energy, where sensor data needs to be collected, monitored, and acted upon in real time (eG Innovations, 'Top Google Cloud Platform (GCP) Services Explained with Use Cases', 2025) [5].

- **Software Development and DevOps:**
  Google Cloud supports modern DevOps practices through tools such as Cloud Build (CI/CD), Cloud Source Repositories (version control), and Artifact Registry (package management). These tools help streamline development pipelines and increase team productivity (Google Cloud, 'Google Cloud Overview', 2025) [2].

3. How Cloud Computing Works

Cloud computing works by delivering computing resources like storage, processing power, and software over the internet from remote data centers. In Google Cloud's case, when a user makes

a request, it's sent through Google's private global fiber network to the nearest data center. There, virtual machines and containers handle the workload using tools like Kubernetes for scaling and load balancing. Users access these resources through web interfaces or APIs and are only billed for what they use. Google Cloud also ensures security by encrypting data in transit and at rest and maintains high reliability through built-in redundancy and automatic failover systems (HowStuffWorks, 'How the Google Cloud Works', 2025) [6].

4. Cloud Computing Services Offered by Google Cloud

Google Cloud provides a wide range of services to support different IT needs:

- **Compute Services:** Run apps using virtual machines (Compute Engine), managed platforms (App Engine), or event-driven code (Cloud Functions) (Google Cloud, 'Google Cloud Platform Services Summary', 2025) [7].

- **Storage and Databases**: Store and manage data using Cloud Storage for files and services like Firestore, Cloud SQL, and Bigtable for different database needs (Google Cloud, 'Google Cloud Platform Services Summary', 2025) [7].

- **Networking:** Enhance performance with Virtual Private Cloud (VPC), load balancing, domain name services (Cloud DNS), and content delivery via Cloud CDN (Google Cloud, 'Google Cloud Platform Services Summary', 2025) [7].

- **Big Data and AI:** Analyze large datasets with BigQuery and build machine learning models using Vertex AI (Google Cloud, 'Google Cloud Platform Services Summary', 2025) [7].

- **Security and Identity**: Use IAM to manage access, Cloud Armor for protection against attacks, and Security Command Center for threat monitoring (Google Cloud, 'Google Cloud Platform Services Summary', 2025) [7].

- **Development Tools**: Automate and manage code with Cloud Build, use Cloud Shell for browser-based commands, and monitor systems with logging tools (Google Cloud, 'Google Cloud Platform Services Summary', 2025) [7].

5. How Google Cloud Differs from Other Cloud Providers

Google Cloud distinguishes itself from other cloud providers in several significant ways:

1. **Data Analytics and Machine Learning**: Google Cloud excels in data analytics, particularly through BigQuery, a fully managed data warehouse. BigQuery allows organizations to run extremely fast SQL queries on large datasets, making it a powerful tool for big data analytics. This gives Google Cloud a competitive edge for businesses dealing with massive data workloads (Pluralsight, 'What is Google Cloud Platform (GCP)?', 2025) [3].

2**. Performance and Network Infrastructure**: A standout feature of Google Cloud is its global fiber network, which is private and owned by Google. This reduces latency and enhances throughput compared to typical internet routing, providing superior performance. It is especially beneficial for real-time applications such as live video streaming or interactive cloud gaming, where low latency is critical (Google Cloud, 'What is Cloud Computing?', 2025) [1].

3. **Support for Open Source and Hybrid Cloud Environments**: Google Cloud is a leader in open-source technologies and hybrid cloud environments. As the original creator of Kubernetes, Google offers extensive support for containerized applications. Additionally, Anthos, a Google Cloud tool, enables the management of applications across both on-premises data centers and other cloud environments. This flexibility makes it an ideal choice for businesses adopting multi-cloud or hybrid strategies (Google Cloud, 'What is Cloud Computing?', 2025) [1].

4. **Pricing and Cost Efficiency:** Google Cloud provides sustained use discounts and per-second billing, which makes it particularly cost-effective for long-running workloads. These pricing models allow businesses to optimize costs based on their usage patterns (Google Cloud, 'Google Cloud Overview', 2025) [2].

5. User-Friendly Interface: Google Cloud's clean, intuitive interface makes it easy for developers and small teams to start using cloud technologies. Its user-centric design is ideal for those new to cloud computing, simplifying the deployment and management of cloud-based applications (Pluralsight, 'What is Google Cloud Platform (GCP)?', 2025) [3].

6. Case Study

Problem:
Spotify had a hard time handling a lot of user data. Their old system was slow, costly, and hard to manage.

Solution:
Spotify moved to Google Cloud. They used tools like BigQuery and Cloud Storage to store and analyze data faster and more easily.

Result:
Spotify could now give better music suggestions, save money, and grow quickly around the world.

Conclusion

In conclusion, this project successfully met its objectives by designing and implementing a structured network tailored for the university's student club environment. Through the use of Cisco Packet Tracer, we developed a reliable system that enhances communication, improves coordination, and simplifies management within the club network. The setup provides better organization, greater control, and ease of use for members. It also lays the groundwork for future expansion and supports ongoing digital transformation. Overall, the project addressed existing challenges and established a solid foundation for efficient and modern club operations.

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